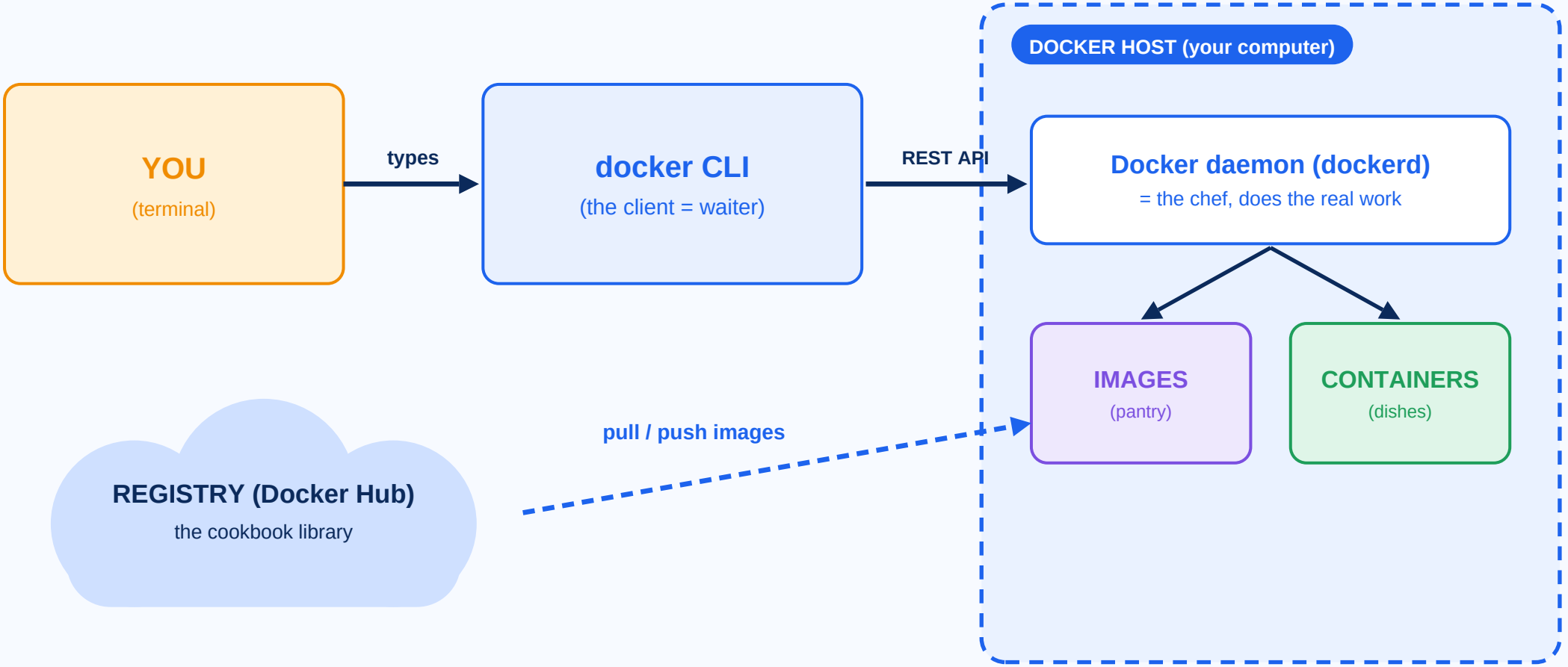
# 1. How Docker Works: Client, Daemon, Registry

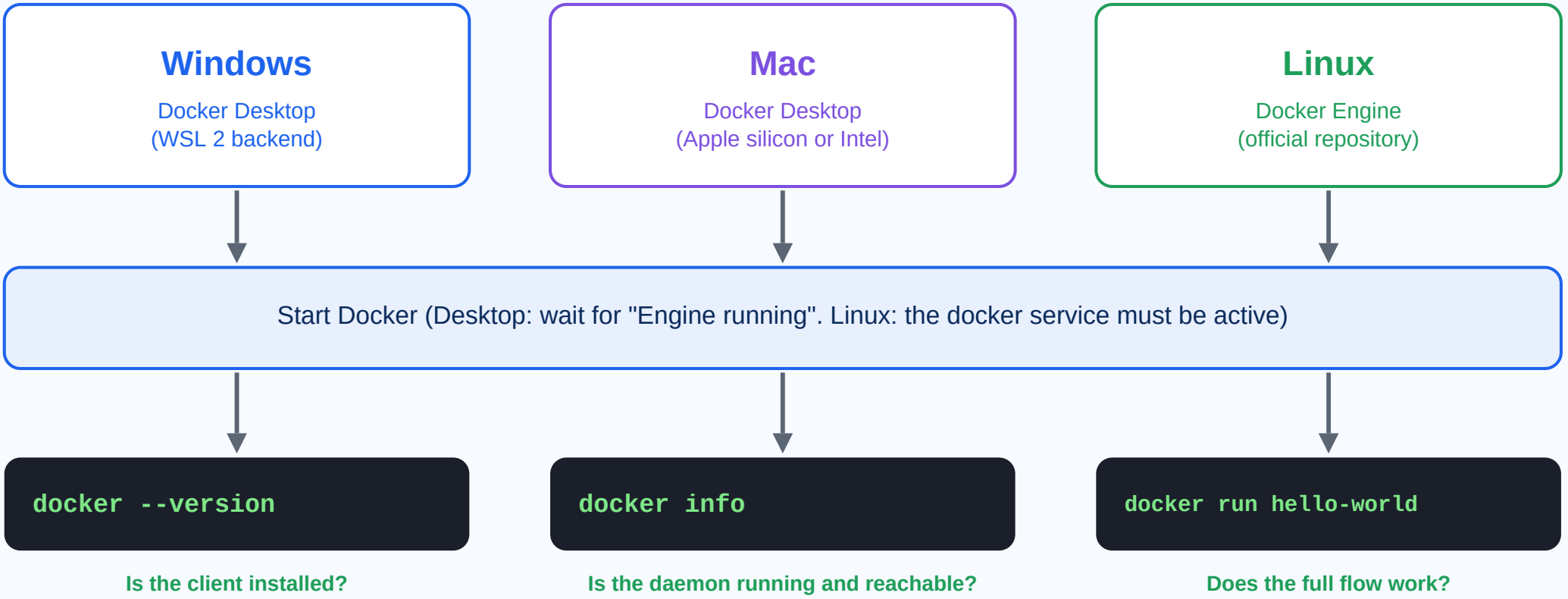
You type commands to the client. The daemon does the work.




Docker Desktop (Windows/Mac) bundles the CLI, the daemon and extra tools in one app. On Linux you install Docker Engine.

# 2. Install and Verify Docker

Pick your system, install, then run the three checks




Linux tip: if you see "permission denied", add your user to the docker group with: sudo usermod -aG docker $USER then log out and back in. Always follow the official guide at docs.docker.com for your exact system.

# 3. Anatomy of a docker run Command

Read it left to right, one piece at a time

```
docker run -d -p 8080:80 --name web nginx
```

**docker**
The tool

**run**
Create + start a container

**-d**
Detached: run in the background

**-p 8080:80**
Publish port: host 8080 to container 80

**--name web**
Give it a name

**nginx**
Which image to use

**General shape: docker run [OPTIONS] IMAGE [COMMAND]**
Options come BEFORE the image name. Anything after the image is a command run inside the container.

**-it**
interactive terminal

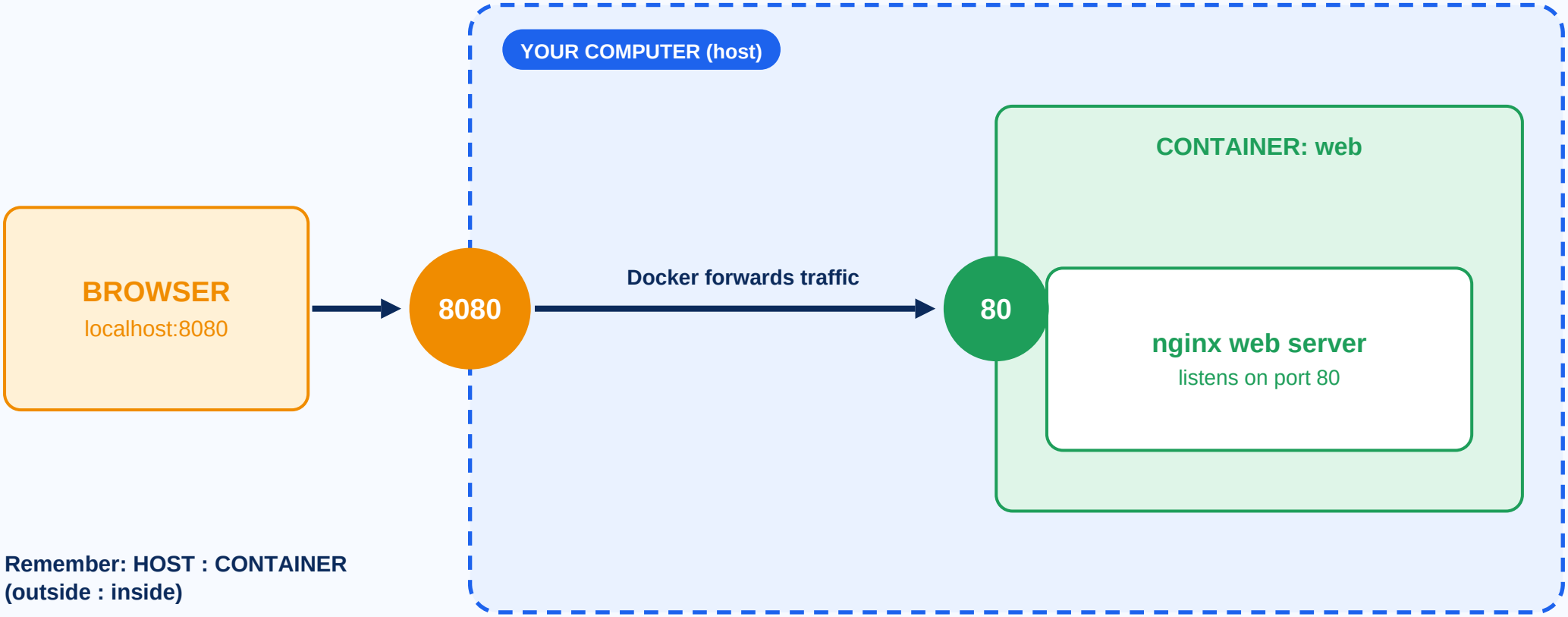
**--rm**
delete when it exits

**-e KEY=val**
set variable (Ep 6)

**-v a:b**
mount storage (Ep 4)

# 4. Port Mapping: -p HOST:CONTAINER

Why localhost:8080 shows the app that listens on port 80 inside




**Wrong order = nothing works**

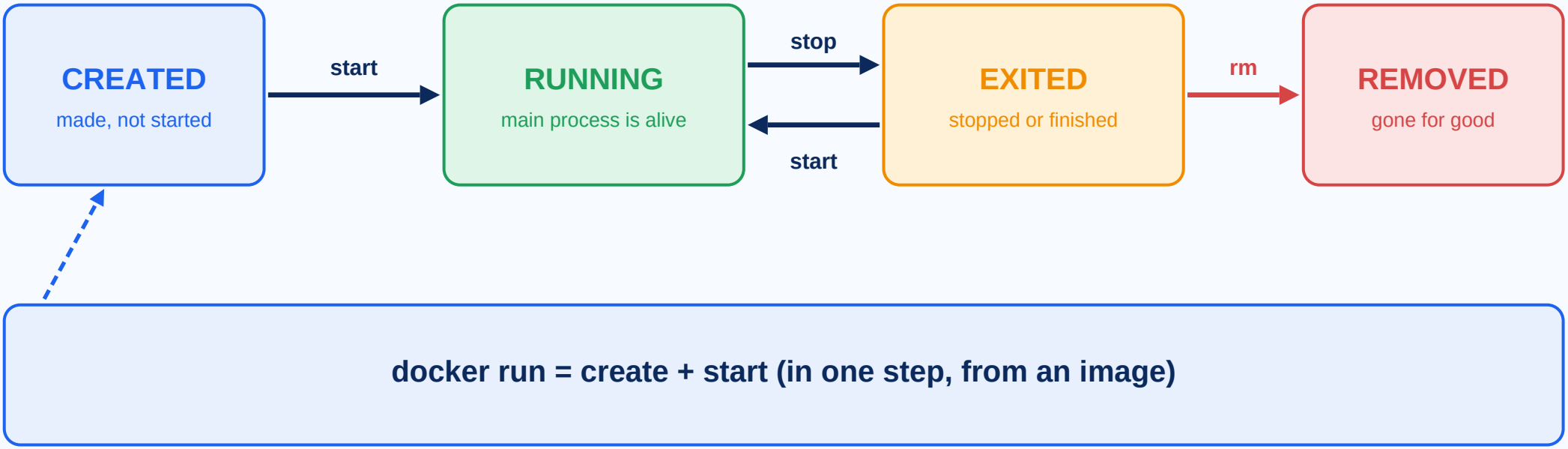
-p 80:8080 would send host port 80 to container port 8080

**Run two web servers side by side**

-p 8080:80 and -p 8081:80 (different host ports)

# 5. The Container Lifecycle

Every container is in one of these states




**A container lives only while its main process runs. If the process ends, the container goes to Exited.**

```
docker ps
```

shows RUNNING containers

```
docker ps -a
```

shows ALL containers

```
docker rm -f web
```

stop + remove in one go

# 6. Command Map: What to Use When

Four groups cover almost everything in this episode

## IMAGES

get, list, delete recipe kits

```
docker pull nginx
```

```
docker images
```

```
docker rmi nginx
```

## RUN

create and control containers

```
docker run -d -p 8080:80 --name web nginx
```

```
docker start / stop / restart web
```

## LOOK

check, debug, step inside

```
docker ps  /  docker ps -a
```

```
docker logs -f web
```

```
docker inspect web
```

```
docker exec -it web bash
```

## CLEAN UP

remove what you do not need

```
docker rm web
```

```
docker rm -f web
```

```
docker container prune
```

# 7. Today's Lab: One Flow, Seven Steps

Run these in order and watch each result

1. **Pull**

```
docker pull nginx
```

2. **Run**

```
docker run -d -p 8080:80 --name web nginx
```

3. **Browse**

```
open localhost:8080
```

You should see
"Welcome to nginx!" in the browser at step 3

4. **Logs**

```
docker logs web
```

In step 4 you will see
request lines from your own visit

5. **Exec**

```
docker exec -it web bash
```

In step 5 type exit
to leave the container

6. **Stop**

```
docker stop web
```

7. **Remove**

```
docker rm web
```